

Title : Efficacy and safety of drug therapy aimed at reducing alcohol consumption in the treatment of alcohol dependence or alcohol use disorders: a systematic review and network meta-analysis protocol

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Author contributions:

K.K. is the guarantor and drafted the manuscript.

All authors contributed to the development of the selection criteria, the risk of bias assessment strategy and data extraction criteria. K.K., T.H. and Y.H. developed the search strategy. M.T. and R.S. provided expertise on Alcohol dependence or alcohol use disorders (AUD). M.B. and Y.K. were involved in network meta-analysis. All authors read, provided feedback, and approved the final manuscript.

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1. Introduction

Excessive alcohol consumption is a social problem (1). According to the World Health Organization, approximately three million deaths were attributed to alcohol consumption, which is 5.3% of all deaths. Despite this high mortality, only a small number of alcoholics receive alcoholism treatment, and many do not receive it (2). The reason for the lack of progress in the treatment of alcohol dependence or AUD has been that the goal of sobriety is a hurdle for alcoholics. Abstinence is a common therapeutic goal, but the perception that abstinence is the only treatment goal may discourage treatment-seeking (3, 4). Recently, it was reported that the prognosis at one year was the same for both abstinence and moderation use goals in the treatment of alcohol dependence and that it was important for patients to choose their own treatment goal of abstinence or reduction in alcohol consumption to continue treatment (5). Therefore, medication to reduce drinking is used for patients whose treatment goal is to reduce drinking. In the past, a network meta-analysis (NMA) of pharmacologically controlled drinking in the treatment of alcoholism or AUD has been reported (6). The review summarizes existing data from randomized controlled trials up to June 2016. However, in recent years, there have been several reports that there are some drugs that can be expected to reduce alcohol consumption including anticonvulsants (7, 8), and some RCTs of drugs to reduce alcohol consumption have been reported (9, 10). This systematic review and NMA aimed to summarize and critically appraise all existing evidence on the efficacy and safety of drug therapy for reducing alcohol consumption in the treatment of alcohol dependence or AUD.

2. Research question

P: Alcohol dependence or AUD

I: Nalmefene, Naltrexone, Topiramate, Baclofen, Carbamazepine, Oxcarbazepine, Levetiracetam, Zonisamide, Valproate, Gabapentin Pregabalin, Ondansetron, Varenicline, Acamprosate

C: Placebo

- O:
1. Change from baseline in heavy drinking days (HDD) (days when daily alcohol intake exceeds 60g for men and 40g for women) (long term after 24 weeks)
 2. Change from baseline in total daily alcohol intake (long term after 24 weeks)
 3. Any adverse events (long term after 24 weeks)

3. Method

3.1 Protocol

We used a systematic review protocol template (11). We searched for studies and presented the results according to the Cochrane Handbook for Systematic Reviews of Interventions (12) and PRISMA statement (13). We will publish this protocol in protocols.io.

3.2 Inclusion criteria of the articles for the review

3.2.1 Type of studies

We will include randomized controlled trials that assess AUD or alcohol dependence. We will only include monotherapy and exclude combination therapy.

We will exclude quasi-randomized controlled trials (such as those allocated by using alternate days of the week) because they can cause selection bias in allocation concealment.

In cross-over trials, we will consider only results from the first phase, that is, before the first cross-over. The expected and unprecedented pharmacological, physiological, and psychological effects of first-stage study interventions can affect the results of second-stage studies in crossover studies.

In cluster-randomized trials, we will make an adjustment to the sample size for each intervention based on the method described in the Cochrane Handbook for Systematic Reviews of Interventions (12), using an estimate of the intraclass correlation coefficient (ICC) derived from the study (where available) or a similar study or a study of a similar population.

We will not apply language or country restrictions. We will include all papers including published, unpublished articles, abstract of conference and letter. We will not exclude studies based on the observation period or publication year.

3.2.2 Study participants

- Alcohol dependence or AUD

Inclusion criteria:

- Over 18 years old
- Both sexes
- All studies are based on each of the following diagnostic criteria for Alcohol dependence or AUD
 - DSM-III (14)
 - DSM- III-R (15)
 - DSM-IV (16)
 - DSM-IV-TR (17)
 - DSM-5 (18)

- ICD-10 (19)
- ICD-11 (20)

Exclusion criteria:

- We will exclude studies where drugs was used to treat alcohol withdrawal.

3.2.3 Intervention

Nalmefene : 10~40mg/day (po) (as needed)

Naltrexone : polar 50mg/day (Maximum 150mg/day) or injectable (iv) from 75 to 400 mg/day

In the case of naltrexone, 50 mg (oral) and 380 mg (injection), for which clinical efficacy was obtained, were considered identical and converted (21).

Topiramate : 25~300mg/day (po)

Baclofen : 30~60mg/day (po)

Carbamazepine : 200~800mg/day (po)

Oxcarbazepine : 600~1800mg/day (po)

Levetiracetam : 500~2000mg (po)

Zonisamide : ~4000mg/day (po)

Valproate : ~2000mg/day (po)

Gabapentin : 900~1800mg/day (po)

Pregabalin : 150~450mg/day (po)

Ondansetron : 1,4,16µg/kg twice per day (po)

Varenicline : 0.5~2mg/day (po)

Acamprosate : 1~3g/day (po)

3.2.4 Control

Placebo

3.3 Type of outcomes

3.3.1 Primary outcomes

1. Change from baseline in heavy drinking days

Definition: (HDD ; a day with a high consumption level)

Period: long term after 24weeks

2. Change from baseline in total daily alcohol intake

Period: long term after 24 weeks

3. Any adverse events

Period: long term after 24 weeks

Definition: Treatment-emergent adverse events

3.3.2 Secondary outcomes

1. Change from baseline in heavy drinking days (HDD)

Definition: (HDD ; a day with a high consumption level)

Period: short term up to 12 weeks : range 4-23 weeks

2. Change from baseline in total daily alcohol intake

Period: short term up to 12 weeks : range 4-23 weeks

3. Any adverse events

Period: short term up to 12 weeks : range 4-23 weeks

Definition: Treatment-emergent adverse events or the most common adverse event reported

4. Change from baseline in days without drinking

Period: long term after 24 weeks

5. Change from baseline in days without drinking

Period: short term up to 12 weeks : range 4-23 weeks

3.4 Search method

3.4.1 Electronic search

We will search the following databases:

1. The Cochrane Central Register of Controlled Trials (CENTRAL);
2. MEDLINE via PubMed;
3. EMBASE via ProQuest Dialog;
4. PsycINFO via PsycNet

See Appendix 1, 2, and 3 for the search strategies.

3.4.2 Other resources

We will also search the following databases for ongoing or unpublished trials:

1. The World Health Organization International Clinical Trials Platform Search Portal (ICTRP);
2. ClinicalTrials.gov;

We will check the reference lists of studies, including international guidelines the World Federation of Societies of Biological Psychiatry (22) and Characteristics of American Psychiatric Association (APA) Clinical (23) and National Institute for Health and Care Excellence (24) and British Association for Psychopharmacology (25) and Canadian Guideline (26) and Australian government department of health (27) as well as the reference lists of eligible studies and articles citing eligible studies. We will ask the authors of original studies for unpublished or additional data.

3.5 Data collection and analysis

3.5.1 Selection of the studies

Three review authors (KK, TH and YH) will screen titles and abstracts, followed by the assessment of the eligibility based on the full texts. We will contact the original authors if relevant data is missing. Disagreements between the three authors will be resolved by discussion, and if this fails, another reviewer will act as an arbiter (MT).

	Search period in Palpacuer's review	Search period that is not included in Palpacuer's review
Nalmefene Naltrexone (po) Topiramate Baclofen Acamprosate	The Cochrane library : April 2015 Embase : April 2015 PubMed/Medline : June 2016 We find the following statement in Palpacuer's review. (6) “Eligible studies were identified from PubMed/Medline, the Cochrane library and Embase, including congress abstracts. The searches were conducted in April 2015. An update of the search was performed for the Pubmed database in June 2016.”	The Cochrane library : May 2015 Embase : May 2015 PubMed/Medline : July 2016
Naltrexone injectable (iv) Carbamazepine Oxcarbazepine Levetiracetam Zonisamide	We will conduct a new search.	We will conduct a new search.

Valproate Gabapentin Pregabalin Ondansetron Varenicline		
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3.5.2 Data extraction and management

Two of three review authors (KK, TH and YH) will independently extract each article/ study report, evaluate the completeness of the data abstraction and confirm the quality rating. We will use data extraction forms to ensure the consistency of information and evaluation of each study (Table 1).

We will extract included study characteristics (such as lead author, publication year and journal), participant characteristics (such as diagnosis of alcohol dependence or AUD, age, sex), intervention details (such as drug dose and duration of drug administration), and outcome measures.

Two of three review authors (KK, TH and YH) verify that the data entered for each other is correct.

When published and unpublished studies provide different values, we will prioritize the unpublished data.

Any disagreements will be resolved by discussion, and if this fails, a third reviewer will act as another arbiter (MT).

	Search period in Palpacuer's review	Search period that is not included in Palpacuer's review
Nalmefene Naltrexone (po) Topiramate Baclofen Accamprosate	Two of three review authors (KK, TH and YH) will independently extract each article/ study report, evaluate the completeness of the data abstraction and confirm the quality rating.	Two of three review authors (KK, TH and YH) will independently extract each article/ study report, evaluate the completeness of the data abstraction and confirm the quality rating.
Naltrexone injectable (iv) Carbamazepine Oxcarbazepine Levetiracetam Zonisamide Valproate Gabapentin Pregabalin Ondansetron Varenicline	Two of three review authors (KK, TH and YH) will independently extract each article/ study report, evaluate the completeness of the data abstraction and confirm the quality rating.	Two of three review authors (KK, TH and YH) will independently extract each article/ study report, evaluate the completeness of the data abstraction and confirm the quality rating.

3.6 Assessment of risk of bias in included studies

Two of three review authors (KK, TH and YH) will evaluate the risk of bias independently using the Risk of Bias 2.0 (28) Disagreements between the two reviewers will be discussed, and if this fails, another reviewer (MT) will be acting as an arbiter. We will assess the following domains:

- Risk of bias arising from the randomization process,
- Risk of bias due to deviations from the intended interventions,
- Missing outcome data
- Risk of bias in the measurement of the outcome
- Risk of bias in the selection of the reported result
- Overall risk of bias

We will judge the level of risk of bias explicitly for each domain as being at 'low', 'high', or 'some concern' risk of bias. Afterward, we will present the results obtained in the "Risk of Bias" table.

3.7 Measures of treatment effects

We will calculate continuous outcomes (change from baseline in heavy drinking days, total daily alcohol intake and days without drinking) by finding the mean difference of 95% confidence intervals (CIs). If the SD of the change is not reported, we will impute the SD at the difference between the start and end of treatment. If that is also difficult, we will ask the author to provide the data.

For dichotomous (any adverse events), we will synthesize risk ratios (RRs) with 95% CIs.

3.8 Unit of analysis issues

Cluster-randomized trials,

We will make an adjustment to the sample size for each intervention based on the method described in the Cochrane Handbook for Systematic Reviews of Interventions (12), using an estimate of the intraclass correlation coefficient (ICC) derived from the study (where available) or a similar study or a study of a similar population.

Cross-over trials

We will consider only results from the first phase, that is, before the first cross-over. The expected and unprecedented pharmacological, physiological, and psychological effects of first-stage study interventions can affect the results of second-stage studies in crossover studies.

Quasi-randomized controlled trials

We will exclude quasi-randomized controlled trials (such as those allocated by using alternate days of the week) because they can cause selection bias in allocation concealment.

3.9 Handling of missing data

To assess the impact of missing outcome data, we will consider the following factors;

Level of missing data: Outcomes of patients who leave the study early are typically imputed by the trialists, often using Last Observation Carried Forward (LOCF) (29). The appropriateness of the imputation method to account for early dropouts will be considered in the Risk of bias assessment section. The extent of missing data will be explicitly reported in the "Characteristics of Included Studies" section. If mixed methods or area-under-the-curve methods are reported, the results of these analyses will be extracted (30).

Differences between groups: In the comparative study of placebo and drugs aimed to reduce alcohol consumption, no significant difference in the number of dropouts was observed when the goal was short-term treatment (31, 32), but a difference in the number of dropouts may occur when the goal was long-term treatment (33). Therefore, in the case of comparison of trials targeting long-term treatment, it is expected to be a problem to compensate for missing data with LOCF.

Reasons for missing data: Inform the authors of the reasons for missing data whenever possible. If data are believed to be missing at random, only available data will be analyzed.

3.10 Assessment of heterogeneity

Assessment of heterogeneity in Pair-wise meta-analysis

A survey of the literature shows that there are not many direct comparative studies comparing the efficacy and safety of drug therapy aimed at reducing alcohol consumption (34-39). Therefore, pair-wise meta-analysis assumes a common heteroskedasticity variance for all direct comparisons. On the other hand, NMA assumes a common estimate of heteroskedasticity variance for different comparisons.

Assessment of heterogeneity in network meta-analysis

The assessment of statistical heterogeneity in the network will be based on the magnitude of the heterogeneity variance parameter (T^2) estimated from the NMA models. For dichotomous outcomes, the magnitude of the heterogeneity variance will be compared with the empirical distribution (40). We will also estimate a total I^2 value for heterogeneity in the network (41).

3.11 Assessment of transitivity assumption

Transitivity assumptions will be assessed by comparing clinical and methodological characteristics (potential effect modifiers presented in data extraction and management). Clinical characteristics that influence the effect of alcohol reduction medications in alcoholism include the severity of dependence, comorbid with psychiatric disorders and daily alcohol consumption at baseline. These variables will be examined to see if they are similarly distributed in studies grouped for comparison.

Data synthesis

3.12 Direct treatment comparisons

When we synthesize studies that compare the same interventions, we will use a random-effects model to perform a pairwise meta-analysis.

3.13 Network meta-analysis

NMA is a method that integrates comparison results based on multiple studies on results such as efficacy and safety of drugs and enables direct and indirect comparisons without direct comparison. NMA combines direct and indirect evidence across a network of randomized trials into a single effect size, and, under certain assumptions, it can increase the precision of the estimates while respecting randomization. This model allows us to estimate the probability that each intervention is optimal for each outcome, given the relative effect sizes estimated by the NMA. As described in the section of assessment of heterogeneity in network meta-analysis, a common heterogeneity variance parameter (T^2) will be assumed for each network of interventions. The results will be presented in a league table with effect sizes and 95% CIs, and the confidence level of the evidence for each comparison will be indicated by a color. We will also use prediction intervals to assess the impact of heterogeneity on the results. To rank the ratio of the effect of the ideal treatment to the effect of the treatment, we will use the surface under the cumulative rank curve (SUCRA); a large SUCRA means a large effect of the treatment (42). We will report the findings for these interventions in the results and the conclusions. We will perform NMA for each primary and secondary outcome measure, using MetaInsight (43). We will carry out analyses for statistical evaluation of the inconsistency and production of network graphs and result figures

3.14 Assessment of inconsistency

If we find a significant inconsistency, we will investigate possible sources. In concrete terms, we will plan to investigate the distribution of pre-specified clinical and methodological variables in each comparator study that we believe may be sources of heterogeneity and inconsistency.

If sufficient studies are available, we will consider the following potential sources of heterogeneity and network inconsistency:

- Presence or absence of physical disease (presence or absence of alcoholic liver disorder)
- Presence or absence of mental illness (schizophrenia, depression, addiction other than alcoholism, dementia, behavioral psychological symptoms of dementia (BPSD), etc.)
- Target patients (outpatients and inpatients)
- Patients with more severe alcohol dependence and drinking a lot of alcohol (according to WHO criteria) (44)
- Different drug doses and routes of administration

3.15 Assessment of reporting bias

Reporting biases arise when the dissemination of research findings is influenced by the nature and direction of the results (45). We will assess the potential publication bias by visual inspection of the funnel plot. Egger test will be performed as well (46, 47). We will not conduct the test when we find less than 10 trials or trials that have a similar sample size.

3.16 Subgroup analysis and network meta-regression

To elucidate the influence of effect modifiers on results, we will evaluate the subgroup analyses and network meta-regression of the primary outcomes on the following factors when sufficient data are available.

1. (For participants)

- Presence or absence of physical disease (presence or absence of alcoholic liver disorder)
- Presence or absence of mental illness (schizophrenia, depression, addiction other than alcoholism, dementia, BPSD, etc.)
- Target patients (outpatients and inpatients)
- Patient's alcohol consumption (according to WHO criteria) (44)

2. (For intervention) : none

3.17 Sensitivity analysis

We will undertake the following sensitivity analyses for the primary outcomes to assess whether the results of the review are robust to the decisions made during the review process.

1. Exclusion of studies using imputed statistics.
2. Missing participants: verify the robustness of the results by seeking informative missingness odds ratios (45).
3. Only the participants who complete the study with complete data.
4. Summary of findings and assessment of the certainty of the evidence

We used the Confidence in Network Meta-Analysis (CINeMA) tool to assess the reliability of the

NMA evidence (48-50). This tool evaluates the following domains:

- Within study bias (Risk of bias contribution)
- Reporting bias
- Indirectness
- Imprecision
- Heterogeneity
- Incoherence

By using CINeMA tool, we can calculate a contribution matrix that shows how much information each study provides to the results of NMA. This contribution rate matrix is combined with the risk-of-bias rating (high, medium, low) to evaluate the relative effectiveness of each comparison in the network. In evaluating imprecision, heterogeneity, and incoherence, we consider the impact of these components of variability informing clinical decisions. As CINeMA assessment requires some threshold values and evaluation rules to be decided, we will make a decision carefully after much discussion.

5. Conflict of Interest

The authors declare no conflicts of interest.

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Appendix 1: CENTRAL search strategy

1. MeSH descriptor: [Alcoholism] explode all trees
2. ((alcohol*) near (abus* or addict* or dependen* or disorder* or drink* or consumption*)):ti,ab,kw_
(Word variations have been searched)
3. MeSH descriptor: [Alcohol Drinking] explode all trees
4. #1 or #2 or #3
5. (Nalmefene or Naltrexone or Topiramate or Baclofen or Levetiracetam or Zonisamide or Valproate or Gabapentin or Pregabalin or Ondansetron or Varenicline or Acamprosate):ti,ab,kw_
6. #4 and #5

Appendix 2: MEDLINE (via PubMed) search strategy

- #1 Alcoholism [mh] or Alcohol Drinking [mh]
- #2 alcohol [tiab] or drink*[tiab]
- #3 addict*[tiab] or abus* [tiab] or dependen*[tiab] or disorder*[tiab] or consumption* [tiab]
- #4 #2 and #3
- #5 #1 or #4
- #6 Naltrexone[mh] or Topiramate[mh] or Baclofen[mh] or Levetiracetam[mh] or Zonisamide[mh] or Valproate[mh] or Gabapentin[mh] or Pregabalin[mh] or Ondansetron[mh] or Varenicline[mh] or Acamprosate[mh]
- #7 Nalmefene[tiab] or Naltrexone[tiab] or Topiramate[tiab] or Baclofen[tiab] or Levetiracetam[tiab] or Zonisamide[tiab] or Valproate[tiab] or Gabapentin[tiab] or Pregabalin[tiab] or Ondansetron[tiab] or Varenicline[tiab] or Acamprosate[tiab]
- #8 #6 or #7
- #9 randomized controlled trial [pt]
- #10 controlled clinical trial [pt]
- #11 randomized [tiab]
- #12 placebo [tiab]
- #13 drug therapy [sh]
- #14 randomly [tiab]
- #15 trial [tiab]
- #16 groups [tiab]
- #17 #9 or #10 or #11 or #12 or #13 or #14 or #15 or #16
- #18 animals [mh] not humans [mh]
- #19 #17 not #18
- #20 #5 and #8 and #19

Appendix 3: EMBASE (via ProQuest Dialog) search strategy

S1 EMB.EXACT.EXPLODE("alcoholism") OR EMB.EXACT.EXPLODE("drinking behavior")
S2 AB(alcohol) OR TI(alcohol) OR AB(drink*) OR TI(drink*)
S3 AB(addict*) OR TI(addict*) OR AB(abus*) OR TI(abus*) OR AB(dependen*) OR TI(dependen*)
OR AB(disorder*) OR TI(disorder*) OR AB(consumption*) OR TI(consumption*)
S4 S2 AND S3
S5 S1 OR S4
S6 EMB.EXACT.EXPLODE("nalmefene") OR EMB.EXACT.EXPLODE("naltrexone") OR
EMB.EXACT.EXPLODE("topiramate") OR EMB.EXACT.EXPLODE("baclofen") OR
EMB.EXACT.EXPLODE("levetiracetam") OR EMB.EXACT.EXPLODE("zonisamide") OR
EMB.EXACT.EXPLODE("valproic acid") OR EMB.EXACT.EXPLODE("gabapentin") OR
EMB.EXACT.EXPLODE("pregabalin") OR EMB.EXACT.EXPLODE("ondansetron") OR
EMB.EXACT.EXPLODE("varenicline") OR EMB.EXACT.EXPLODE("acamprosate")
S7 AB(nalmefene) OR TI(nalmefene) OR AB(naltrexone) OR TI(naltrexone) OR AB(topiramate) OR
TI(topiramate) OR AB(baclofen) OR TI(baclofen) OR AB(levetiracetam) OR TI(levetiracetam) OR
AB(zonisamide) OR TI(zonisamide) OR AB(valproate) OR TI(valproate) OR AB("valproic acid")
OR TI("valproic acid") OR AB(gabapentin) OR TI(gabapentin) OR AB(pregabalin) OR
TI(pregabalin) OR AB(ondansetron) OR TI(ondansetron) OR AB(varenicline) OR TI(varenicline) OR
AB(acamprosate) OR TI(acamprosate)
S8 S6 OR S7
S9 TI,AB(random*) OR TI,AB,SU(placebo) OR TI,AB("double-blind*")
S10 S5 AND S8 AND S9

Appendix 4: PsycINFO (via PsycNet)

#1 Index Terms: {Alcoholism} OR Index Terms: {Alcohol drinking}
#2 Abstract: alcohol OR Title: alcohol OR Abstract: drink* OR Title: drink*
#3 Abstract: addict* OR Title: addict* OR Abstract: abus* OR Title: abus* OR Abstract: dependen*
OR Title: dependen* OR Abstract: disorder* OR Title: disorder* OR Abstract: consumption* OR
Title: consumption*
#4 #2 AND #3
#5 #1 OR #4
#6 Index Terms: {Naltrexone} OR Index Terms: {Topiramate} OR Index Terms: {Baclofen} OR Index
Terms: {Levetiracetam} OR Index Terms: {Zonisamide} OR Index Terms: {Valproate} OR Index
Terms: {Gabapentin} OR Index Terms: {Pregabalin} OR Index Terms: {Ondansetron} OR Index
Terms: {Varenicline} OR Index Terms: {Acamprosate}
#7 Abstract: Nalmefene OR Title: Nalmefene OR Abstract: Naltrexone OR Title: Naltrexone

Abstract: Topiramate OR Title: Topiramate OR Abstract: Baclofen OR Title: Baclofen OR Abstract:
Levetiracetam OR Title: Levetiracetam OR Abstract: Zonisamide OR Title: Zonisamide
Abstract: Valproate OR Title: Valproate OR Abstract: Gabapentin OR Title: Gabapentin OR Abstract:
Pregabalin OR Title: Pregabalin OR Abstract: Ondansetron OR Title: Ondansetron OR Abstract:
Varenicline OR Title: Varenicline OR Abstract: Acamprosate OR Title: Acamprosate
#8 #6 OR #7
#9 Any Field: "double-blind" OR Any Field: "random* assigned" OR Any Field: control
#10 #5 AND #8 AND #9

Appendix 5: ICTRP search strategy

Condition or disease : Alcoholism or Alcohol Drinking

Intervention : Nalmefene or Naltrexone or Topiramate or Baclofen or Levetiracetam or Zonisamide
or Valproate or Gabapentin or Pregabalin or Ondansetron or Varenicline or Acamprosate

Appendix 6: ClinicalTrials.gov search strategy

Condition or disease : Alcoholism or Alcohol Drinking

Intervention : Nalmefene or Naltrexone or Topiramate or Baclofen or Levetiracetam or Zonisamide
or Valproate or Gabapentin or Pregabalin or Ondansetron or Varenicline or Acamprosate

Study Type : Interventional Studies (Clinical Trials)